

PERSONAL FINANCE ADVISOR BOT

AI-Powered Personal Financial Planning and Budget Management Platform

Powered By	Gemini AI, Flask, SQLAlchemy and SQLite
Core Modules	Income, Expenses, Budget Planner, AI Advisor, Savings & Goals, Reports
Technology	Python, Flask, SQLAlchemy, SQLite, JavaScript, HTML, CSS, Gemini AI
Project Type	AI-Powered Full-Stack Web Application
Mentor	

1. Project Description

The Personal Finance Advisor Bot is an AI-powered financial planning assistant designed to help users manage their personal finances through a centralized digital dashboard. The system allows users to record income, track expenses, monitor monthly budgets, set savings goals, review financial activity, and receive AI-powered financial guidance.

The application presents financial information through clear cards, progress indicators, charts, transaction lists, and financial goals. This reduces the complexity of manual financial tracking and gives users a structured view of their monthly financial activity.

The project is designed as a full-stack web application using Python and Flask for the backend, SQLAlchemy and SQLite for data management, JavaScript for interactive functionality, and Gemini AI for intelligent financial assistance.

2. Project Overview

The Personal Finance Advisor Bot combines common personal finance activities into one dashboard. The interface shown in the project design provides a monthly financial overview with total income, total expenses, net savings, savings rate, budget progress, savings goals, recent transactions, and upcoming financial goals.

The dashboard example uses September 2026 data and demonstrates how a user can view financial information at a glance. The displayed profile includes monthly income of ₹35,000, total expenses of ₹22,450, net savings of ₹12,550, and a savings rate of 35.8%.

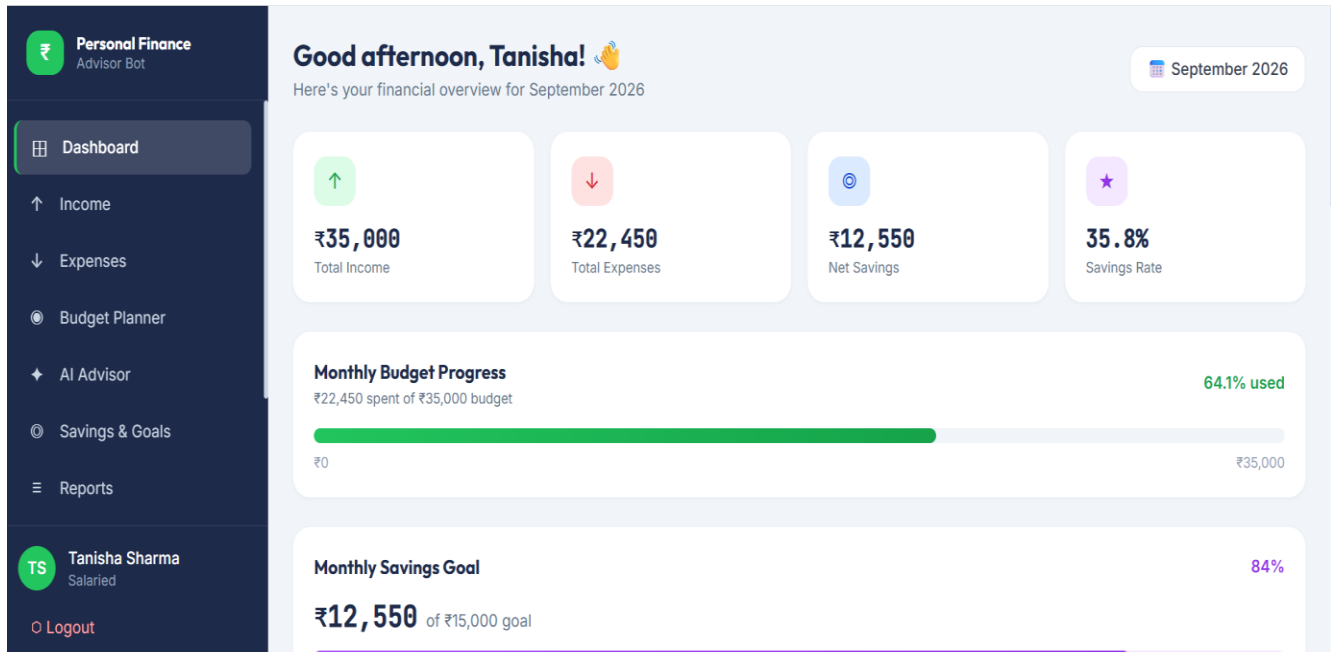


Figure 1: Personal Finance Advisor Bot Dashboard – Monthly Financial Overview

3. Objectives of the Project

Objective 1 – Income Management

To provide users with a simple method for recording and monitoring monthly income.

Objective 2 – Expense Tracking

To allow users to record expenses and organize them into meaningful categories.

Objective 3 – Monthly Budget Monitoring

To help users compare their spending against the monthly budget and monitor budget utilization.

Objective 4 – Savings Goal Management

To allow users to set monthly savings targets and monitor progress toward those targets.

Objective 5 – Expense Analysis

To present expense information in a structured form so users can understand major spending categories.

Objective 6 – Income vs Expenses Analysis

To provide a clear comparison between income and expenses and show the resulting savings.

Objective 7 – AI Financial Guidance

To integrate Gemini AI for conversational financial tips and planning support.

Objective 8 – Financial Goal Tracking

To help users monitor upcoming financial goals such as an emergency fund or planned purchases.

Objective 9 – Transaction Monitoring

To provide a recent transaction view showing income and expense activity.

4. Key Features

4.1 Monthly Budget Progress

The Monthly Budget Progress module shows how much of the monthly budget has been used. In the provided dashboard, ₹22,450 is shown as spent from a ₹35,000 budget, with 64.1% used. A horizontal progress bar provides a visual representation of budget utilization.

4.2 Monthly Savings Goal

The Monthly Savings Goal module allows users to compare current savings with a target. The provided design shows ₹12,550 saved toward a ₹15,000 goal, representing 84% progress.

4.3 Expense Breakdown

The Expense Breakdown module is intended to present category-wise spending so that users can understand where their money is being spent. Categories can include food, transport, entertainment, utilities, rent, education, and other expenses.

4.4 Income vs Expenses

The Income vs Expenses module compares total monthly income with total monthly expenses. The dashboard example shows ₹35,000 total income and ₹22,450 total expenses, leaving ₹12,550 as net savings.

4.5 Recent Transactions

The Recent Transactions section provides a chronological view of financial activity. The supplied design includes transactions such as Monthly Salary (+₹35,000), Metro Card Recharge (-₹500), Netflix Subscription (-₹649), Electricity Bill (-₹1,200), and Freelance Project (+₹8,000).

4.6 Upcoming Financial Goals

The Upcoming Financial Goals section displays future goals with their progress and due dates. The supplied design shows an Emergency Fund with ₹45,000 saved toward a ₹1,00,000 target and a New Laptop goal with 40% progress.

4.7 AI Advisor

The AI Advisor module uses Gemini AI to provide financial guidance based on user questions and relevant financial information entered into the application.

4.8 Dashboard

The dashboard combines financial indicators, budget progress, savings goals, transactions, and financial goals into one centralized interface.

5. Dashboard Figures

5.1 Monthly Budget Progress

The Monthly Budget Progress card visually indicates the amount spent against the monthly budget. The supplied design shows ₹22,450 spent from a ₹35,000 budget and a utilization value of 64.1%.

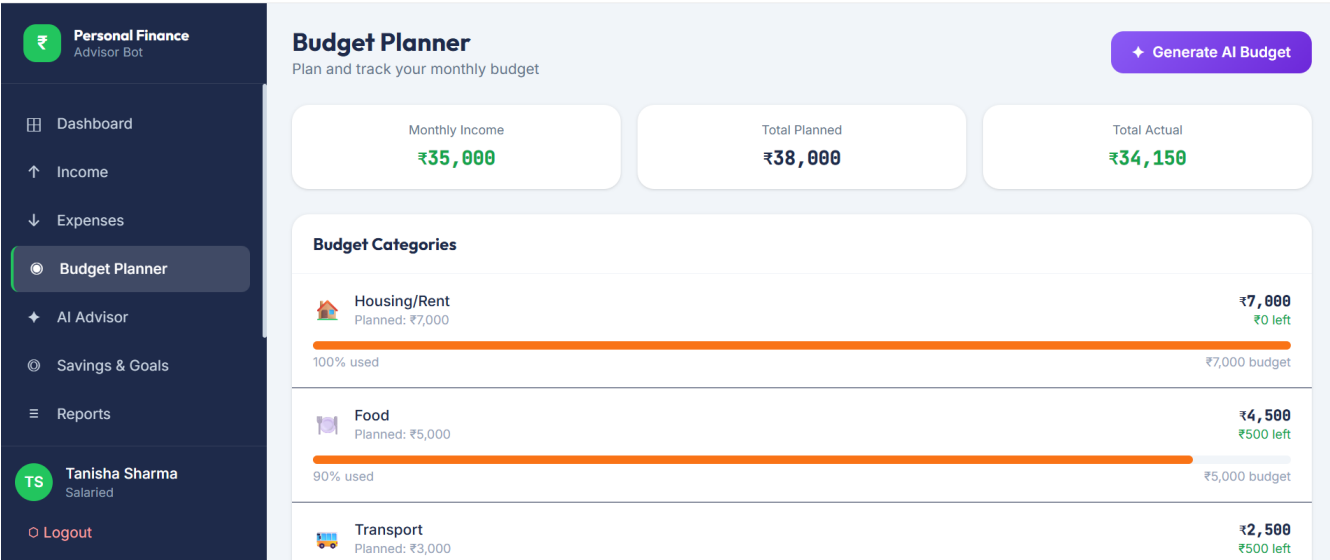


Figure 2: Monthly Budget Progress – ₹22,450 of ₹35,000 budget used

5.2 Monthly Savings Goal

The Monthly Savings Goal card tracks progress toward the monthly savings target. The supplied design displays ₹12,550 saved out of a ₹15,000 goal and 84% progress.

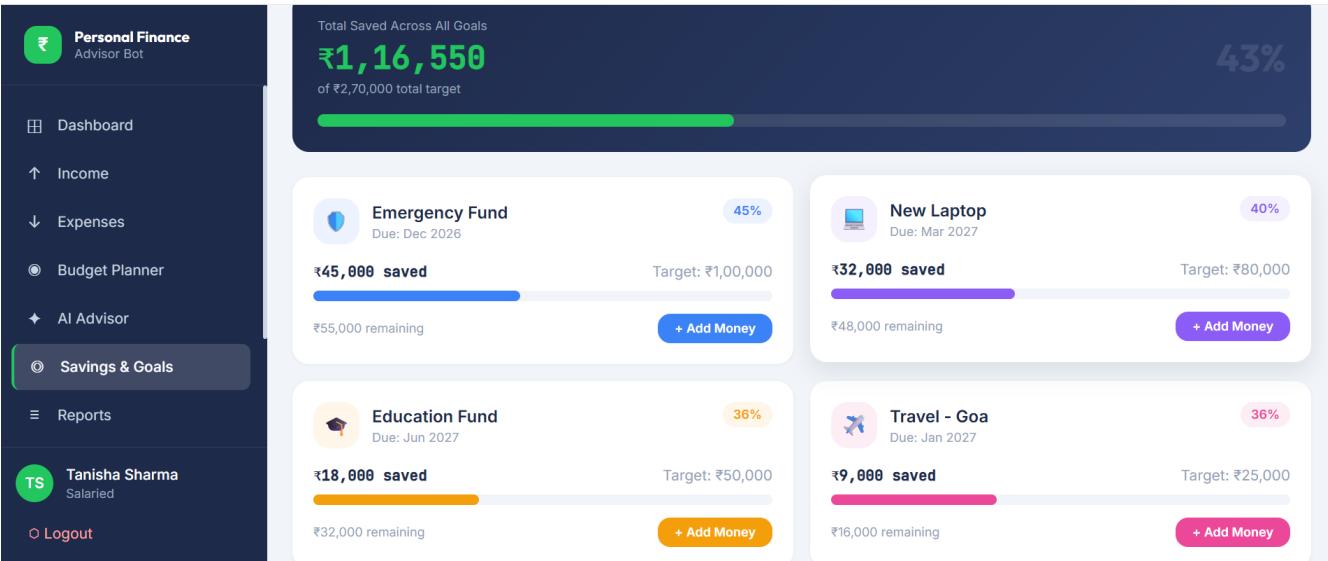


Figure 3: Monthly Savings Goal – ₹12,550 of ₹15,000 goal

5.3 Expense Breakdown

The Expense Breakdown feature provides category-wise visibility into spending. It supports analysis of major expense areas and can be represented using a chart or category summary in the final dashboard.

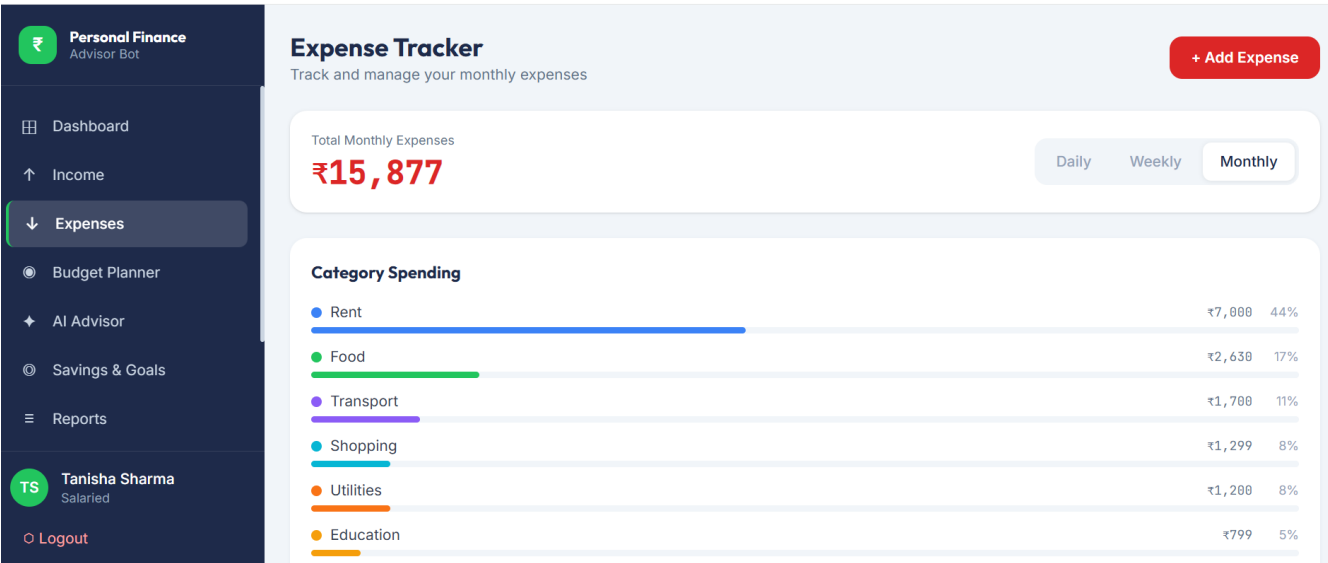


Figure 4: Recent financial activity and dashboard area supporting expense analysis

5.4 Income vs Expenses

The Income vs Expenses view compares the user's monthly income with total spending. In the supplied dashboard, total income is ₹35,000 and total expenses are ₹22,450, resulting in net savings of ₹12,550.

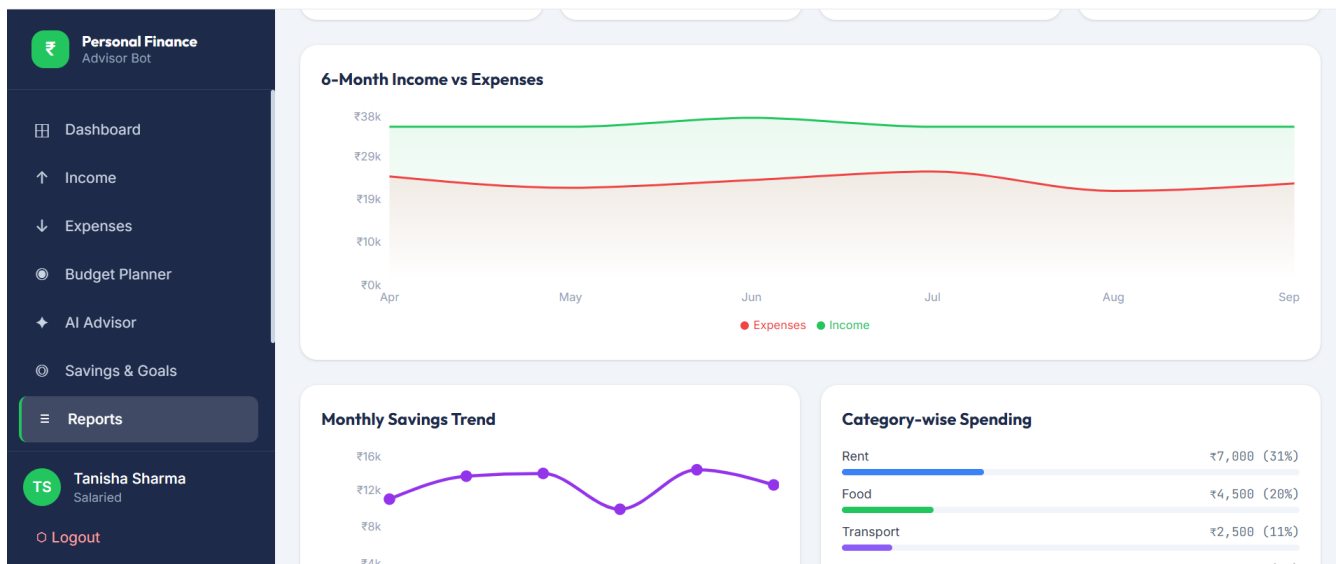


Figure 5: Income, Expenses and Net Savings displayed on the dashboard

5.5 Recent Transactions

The Recent Transactions section provides a list of recent financial entries. Income entries are shown separately from expense entries, helping users quickly identify money received and money spent.

Date	Description	Category	Method	Amount	
22 Sep	Swiggy Order	Food	UPI	-₹480	Delete
20 Sep	Metro Card Recharge	Transport	UPI	-₹500	Delete
19 Sep	Netflix Subscription	Entertainment	Credit Card	-₹649	Delete
18 Sep	Electricity Bill	Utilities	UPI	-₹1,200	Delete
16 Sep	Grocery - BigBasket	Food	UPI	-₹2,150	Delete
15 Sep	Amazon Purchase	Shopping	Credit Card	-₹1,299	Delete
14 Sep	Course Fee - Udemy	Education	Credit Card	-₹799	Delete
13 Sep	Petrol	Transport	UPI	-₹1,200	Delete
12 Sep	Doctor Visit	Healthcare	Cash	-₹600	Delete

Figure 6: Recent Transactions – income and expense activity

5.6 Upcoming Financial Goals

The Upcoming Financial Goals section helps users monitor longer-term targets. The supplied design includes an Emergency Fund and a New Laptop goal with progress percentages and due dates.

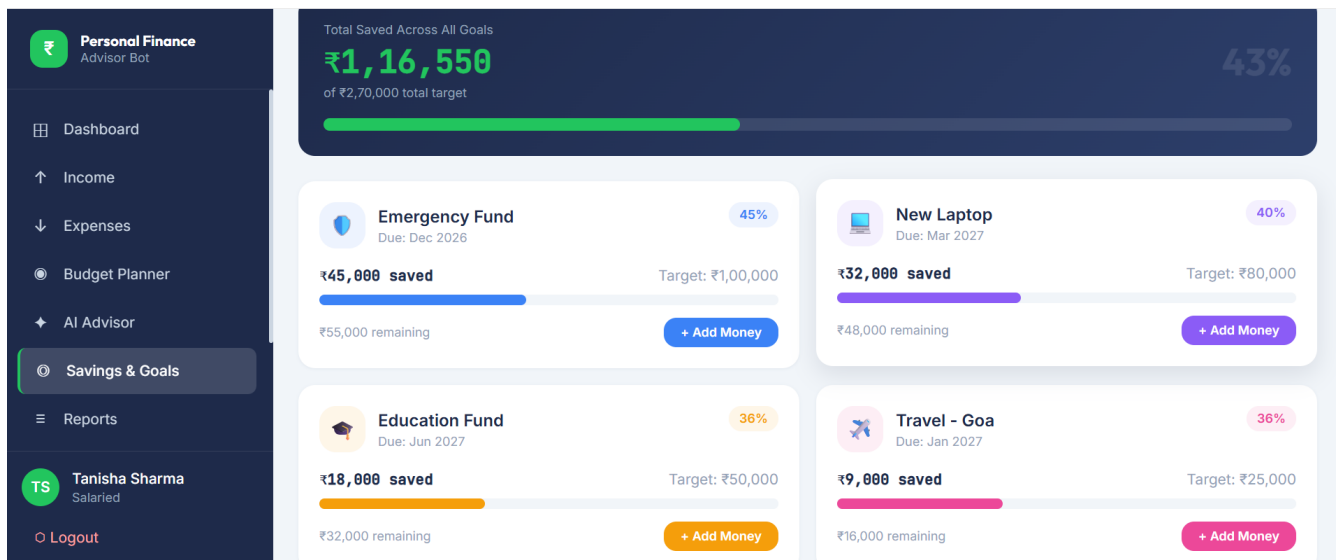


Figure 7: Upcoming Financial Goals

6. Powered By

- Gemini AI – AI-powered financial assistance and guidance.
- Flask – backend web application framework.
- SQLAlchemy – database ORM and data management.
- SQLite – lightweight relational database.
- JavaScript – interactive dashboard and frontend behavior.

7. Core Modules

- Income Management
- Expense Management
- Budget Planner
- AI Advisor
- Savings & Goals
- Financial Reports
- Recent Transactions
- Financial Dashboard

8. Technology

Technology	Purpose	Application
Python	Backend programming	Application logic and financial processing
Flask	Web framework	Routes, APIs and server-side functionality
SQLAlchemy	ORM	Database models and queries
SQLite	Database	Storage of income, expenses, budgets and goals

JavaScript	Frontend programming	Interactive dashboard and dynamic UI
Gemini AI	Artificial Intelligence	AI financial guidance and recommendations
HTML & CSS	Frontend	Page structure and styling

9. Conclusion

The Personal Finance Advisor Bot provides a centralized and user-friendly environment for personal financial management. The supplied dashboard design demonstrates the key information users need for monthly planning, including income, expenses, net savings, savings rate, budget progress, savings goals, recent transactions, and upcoming financial goals.

By combining Flask, SQLAlchemy, SQLite, JavaScript and Gemini AI, the project provides a foundation for intelligent financial tracking and planning. Future versions can extend the platform with more advanced expense analytics, predictive spending insights, additional financial goals, detailed reports, and expanded AI assistance.